Fraud Detection Dataset Report

Data description and data-quality analysis

1. Purpose, Source, Size and Structure

Purpose: The dataset is used to detect fraudulent online card payments. Each row is one transaction, and the goal is to predict whether it is fraud or not.

Source: It looks like the IEEE-CIS Fraud Detection data (Kaggle), provided by the payment company Vesta. Some values are masked to protect privacy.

Size and structure: The data comes in two tables that are joined with `TransactionID`.

Table	Rows	Columns	Content
Transaction	590,540	394	Payment, card, address, email and engineered features
Identity	144,233	41	Device and browser information (`id_01` – `id_38`, `DeviceType`, `DeviceInfo`)

Only about 24% of transactions (144,233 of 590,540) have identity information.

2. Column Description

Column(s)	Type	Possible values	Meaning
`TransactionID`	int	2,987,004 – 3,577,534	Unique ID of each transaction
`isFraud`	int	0 or 1	**Target.** 1 = fraud, 0 = not fraud
`TransactionDT`	int	86,400 – 15,811,131	Time in seconds from a hidden start date (about 183 days). Not a real date

Column(s)	Type	Possible values	Meaning
`TransactionAmt`	float	0.25 – 31,937.39	Payment amount in USD
`ProductCD`	text	W, C, R, H, S	Product code (W is 74% of rows)
`card1`, `card2`, `card3`, `card5`	number	e.g. card1: 1,000 – 18,396	Masked card details (card ID, bank, country, etc.)
`card4`	text	visa, mastercard, american express, discover	Card network
`card6`	text	debit, credit, debit or credit, charge card	Card type
`addr1`, `addr2`	number	addr1: 100 – 540, addr2: 10 – 102	Billing region and country codes
`dist1`, `dist2`	float	0 – 11,623	Distance between addresses
`P_emaildomain`, `R_emaildomain`	text	gmail.com, yahoo.com, hotmail.com, anonymous.com, etc. (about 60 values)	Email domain of the buyer (P) and the receiver (R)
`C1` – `C14`	float	0 – 5,691	Counts (for example, how many addresses are linked to the card). Real meaning is masked
`D1` – `D15`	float	about -193 – 879	Days between events, such as the previous transaction
`M1` – `M9`	text	T / F (M4: M0, M1, M2)	Match flags, such as name on card matches address
`V1` – `V339`	float	Mostly small counts	Features made by Vesta (rankings,

Column(s)	Type	Possible values	Meaning
			counts, links between entities)
`id_01` – `id_38`	mixed	e.g. id_01: -100 – 0, id_12: Found / NotFound	Identity and network details such as proxy, browser and screen size
`DeviceType`	text	desktop, mobile	Type of device used
`DeviceInfo`	text	1,786 values (Windows, iOS Device, MacOS, phone models)	Device name or browser build

Categorical columns: `ProductCD`, `card1` – `card6`, `addr1`, `addr2`, `P_emaildomain`, `R_emaildomain`, `M1` – `M9`, and most text `id_` columns.

3. Data-Quality Analysis

3.1 Missing values

- The transaction table is only **58.9% complete**: about 95.6 million cells are empty.
- The identity table is **64.4% complete**.
- Only five columns have no missing values: `TransactionID`, `isFraud`, `TransactionDT`, `TransactionAmt` and `ProductCD`.

Table	Columns with the most missing values	Missing
Transaction	`dist2`, `D7`, `D13`, `D14`, `D12`	89% – 94%
Transaction	`M1` – `M9`	29% – 59%
Transaction	`card4`, `card6`	about 0.3%
Identity	`id_21` – `id_27`, `id_07`, `id_08`	about 96%
Identity	`id_18`, `id_04`, `id_03`, `id_33`, `id_30`	46% – 69%

Table	Columns with the most missing values	Missing
Identity	`DeviceInfo`	17.7%

3.2 Invalid or unusual values

- **Negative time values:** `D4` (down to -122), `D14` (-193), `D6`, `D12`, `D15` (-83) and `D11` (-53) have negative values, but a number of days between events should not be negative.
- **Almost constant column:** `M1` has "F" only 25 times and "T" 319,415 times, so it adds very little information.
- **Rare categories:** `card6` has "debit or credit" (30 rows) and "charge card" (15 rows). `id_34` has "match_status:-1" (3 rows).
- **Messy text:** `DeviceInfo` has 1,786 values, `id_31` has 130 browser versions, `id_33` has 260 screen sizes and `id_30` has 75 operating systems. These need cleaning and grouping.
- **Hidden meaning:** Many columns (`C`, `D`, `M`, `V`, `card`, `id_`) are masked, so we cannot check them against real-world rules.

3.3 Outliers

Outliers were found with the IQR method (values far outside the middle 50%).

Column	Range	Share flagged	Comment
`TransactionAmt`	0.25 – 31,937	11.3%	Most payments are under 250 USD, but a few are very large
`C1`, `C2`, `C11`, `C13`, `C14`	0 – 5,691	10% – 16%	Most values are 0–6, with a very long tail
`dist1`	0 – 10,286	6.8%	Very long tail
`D1`, `D10`	up to 876	7% – 13%	Long tail
`id_13`, `id_05`, `id_01`	various	18% – 29%	Many flagged values

Many flagged values are probably real, because the data is very skewed. We should check them before removing anything. `isFraud` also shows as an "outlier" in 3.5% of rows. This is the fraud class, not an error.

4. Identifiers, Target and Useful Attributes

- **Identifier:** `TransactionID` joins the two tables. `TransactionDT` gives the order of events. Combining `card1`–`card6` with `addr1` can help to recognise the same customer.
- **Target:** `isFraud`. About 3.5% of transactions (around 20,663) are fraud, so the classes are **very imbalanced**.
- **Potentially useful attributes:**
  - `TransactionAmt` and `ProductCD`
  - `card1`–`card6` and `addr1`, `addr2`
  - Email domains
  - `C`, `D` and `V` features
  - Device and browser details from the identity table

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5. Main Observations and Issues

1. The dataset is large (590,540 rows, 394 columns) and the target is imbalanced (about 3.5% fraud).
2. Missing data is the biggest problem. Many columns are more than 90% empty, and only 24% of transactions have identity data.
3. Negative values in the `D` columns need checking.
4. Heavy skew and many outliers in amounts, counts and distances. Large values should be kept unless they are clearly wrong.
5. Many columns are masked, so we must rely on statistics, not business meaning.